End to End FDE Project

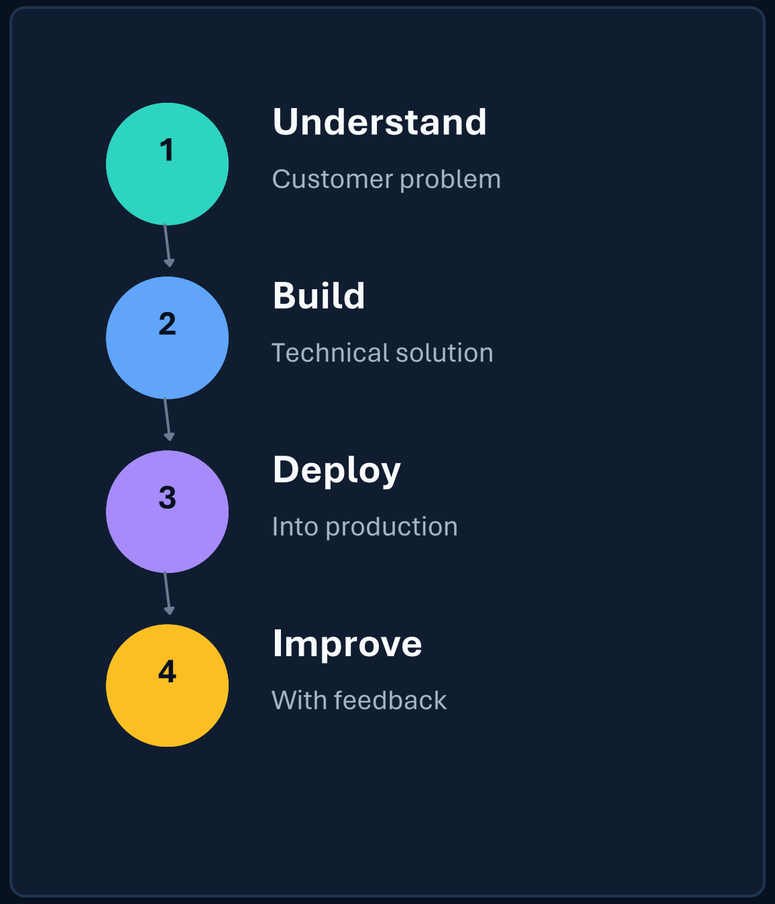
From Customer Problem Statement to Production

Real Problem | Development | Production

FORWARD DEPLOYED ENGINEER

The engineer who turns a customer problem into a working production solution

Customer-facing **Hands-on** **Production**



What is a Forward Deployed Engineer?

Think of an FDE as a bridge between the customer and the technology team.

Customer

“Here is our business problem, data, workflow and constraints.”

FDE

Understands the problem, designs the solution, writes code, integrates systems and gets it running.

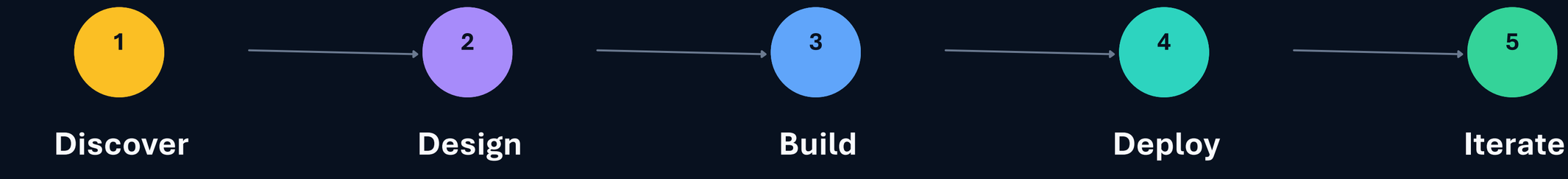
Product / Platform

Provides APIs, AI models, infrastructure, databases and reusable product capabilities.

The FDE translates business needs ↔ technical implementation.

The FDE Mission

Own the journey from “we have a problem” to “it works for real users.”



Key idea: FDEs are measured by customer outcomes—not just by how much code they write.

Core Responsibilities

What does an FDE actually do day to day?

1. Understand requirements

Meet users, ask questions, map workflows, identify pain points.

2. Prototype quickly

Build a small working version to validate the idea.

3. Engineer the solution

Write production code, APIs, data pipelines, AI/RAG/agents, integrations.

4. Deploy & integrate

Connect customer systems, cloud, auth, databases and monitoring.

5. Troubleshoot

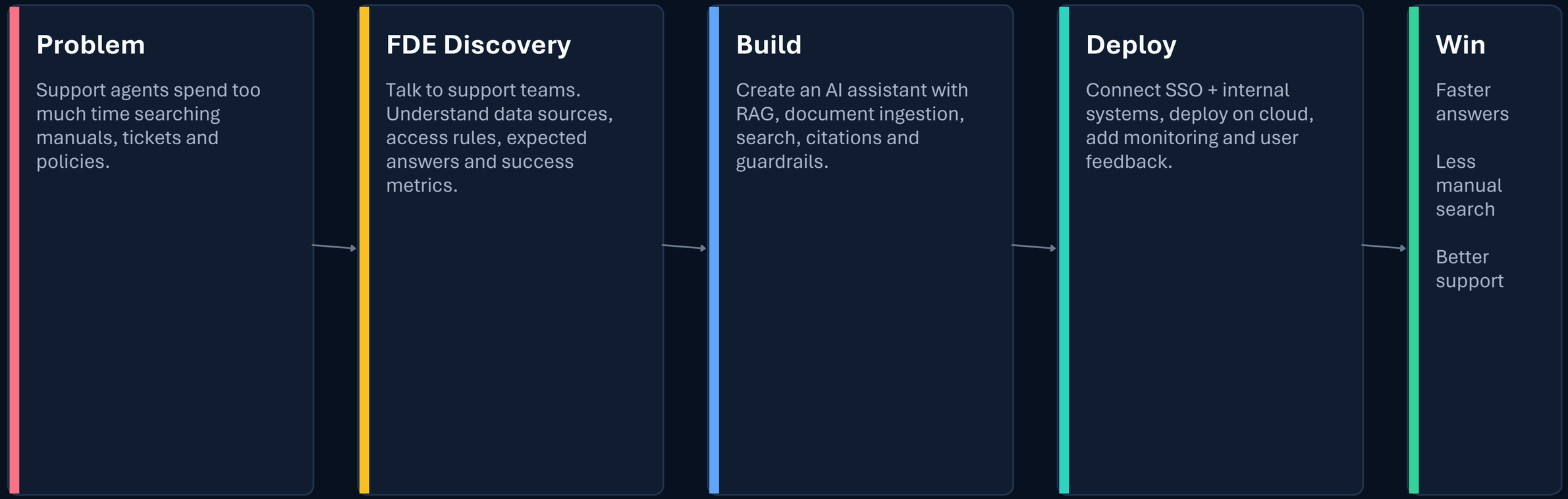
Debug failures, latency, bad outputs, data issues and edge cases.

6. Iterate with users

Collect feedback, measure impact and improve the solution.

A Simple Real-World Example

A support company has 100,000 internal documents. Employees cannot find answers quickly.



FDE success = the customer can actually use it and gets measurable value.

What Makes an FDE Different?

The role combines multiple skill sets in one person.

1. **Software Engineer** — Strong coding, APIs, systems, debugging
2. **AI / Data Engineer** — LLMs, RAG, agents, data pipelines, evaluation
3. **Solution Architect** — Designs how systems connect and scale
4. **Consultant** — Understands business context and asks the right questions
5. **Product Thinker** — Prioritizes user value and fast iteration

FDE = Engineer + Architect + Problem Solver + Customer Partner

FDE vs Traditional Software Engineer

Both are engineers—but their operating environment is different.

Traditional Software Engineer

- Often builds a core product
- Requirements usually come through product teams
- Works mostly inside the engineering organization
- Optimizes reusable product features
- Customer exposure varies

Forward Deployed Engineer

- Builds for a specific customer problem
- Often discovers requirements directly
- Works closely with customer teams
- Optimizes time-to-value + production fit
- High customer interaction

FDE vs Solutions Engineer / Consultant

These roles overlap, but the FDE is typically more implementation-heavy.

Solutions Engineer

Demo + technical pre-sales

Proves the product can solve the problem

Often before / around the sale

Consultant

Advice + transformation

Recommends processes and solutions

May or may not build the system

FDE

Build + integrate + deploy

Owns a working customer outcome

Deep hands-on engineering

Exact titles vary by company. Always read the job description.

Typical AI-FDE Technical Stack

You do not need every tool—but you need strong fundamentals across the stack.

Application	Python • FastAPI • backend services
AI	LLMs • prompting • RAG • agents • tool calling • evaluation
Data	SQL • vector databases • ETL • document pipelines
Integration	REST APIs • webhooks • OAuth/SSO • third-party systems
Cloud / DevOps	Docker • CI/CD • cloud deployment • observability
Engineering	Git • testing • security • debugging • system design

The Most Important Non-Technical Skills

Technical ability gets you in. These skills make you effective with customers.

Communication

Explain complex systems simply.

Problem framing

Find the real problem behind the request.

Ownership

Take responsibility from discovery to production.

Speed

Prototype quickly without losing engineering judgment.

Adaptability

Every customer environment is different.

Trust

Be reliable with customer data, deadlines and expectations.

FORWARD DEPLOYED ENGINEER

Understand → Build → Integrate → Deploy → Improve

The best FDE is not the person who knows every technology.
It is the person who can understand a real problem, choose the right tools, and deliver a solution that works in production.

Customer problem → Production outcome